

Natanel Richey

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Summary

Software-developer skilled in Python and JavaScript focused on Applied Machine Learning and Web/App Design Experience in training AI models, building full-stack websites, data analysis (AI enhanced). Methodical and analytical, hard-working and efficient, fast-learning, hungry for excellence and passionate about bridging the gap between human and machine intelligence.

Professional Experience

Machine Learning Engineer, BabyClue

Nov 2025 - Present

Built and evaluated a multi-class infant-cry classification pipeline, focusing on robust validation, interpretability, and reproducibility.

- Developed end-to-end ML evaluation workflows in Python
- Ran rigorous, comparative model testing using nested CV with fold-level and aggregated analysis (macro/micro F1, per-class precision/recall/F1, confusion matrices, latency) to reliably determine which approaches generalized best across splits
- Built the full training pipeline combining audio preprocessing (spectrogram-based features / AST embeddings), data augmentation, feature construction (late-fusion of embedding + acoustic features), and model training/inference across multiple model families (e.g., MLP, LightGBM, plus classical baselines like SVM/Nearest Centroid where applicable)
- Engineered prompt variants for LLM classification (Chain-of-Thought, Tree-of-Thought variants), systematically compared API responses across providers, evaluating reasoning quality, and refining prompt structure and routing logic to reduce token costs while refining performance

Founder, Even Pinah Services · <https://evenpinah.services/>

June 2025 - Present

Education

BSc. in Computer Science, B.A. in Psychology (Hons.)

2021 - 2025

Hebrew University of Jerusalem, Jerusalem, Israel

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, SQL, C, C++, Java
- **Tools:** Cursor + CLI, MCP, skills.md, Git, Docker, Postman, Cloudinary, Vercel, Weights & Biases
- **AI/ML:** TensorFlow, PyTorch, Scikit-learn, HuggingFace Transformers, LoRA Fine-tuning, Hyperparameter Tuning, RAG implementation, LangChain, LangGraph, Multi-Agent Orchestration, Prompt Engineering, LLM Provider Integration (OpenAI, Anthropic, Google Gemini), Embeddings & Vector Databases (ChromaDB, FAISS), Semantic Caching, Output Validation & Guardrails, A/B Testing for LLMs, NLP, Sentiment Analysis
- **Frontend:** React, Next.js, Tailwind CSS, HTML/CSS, Framer Motion
- **Backend:** FastAPI, Node.js, Express, MongoDB, PostgreSQL, REST APIs, Redis, React Query, pydantic-settings, structlog
- **Other:** OOP, Operating Systems, Algorithms and Data Structures, Statistics and Computational Methods, Data Science, PWA, Web Scraping

Projects

AI Wellness Companion — Production GenAI System

Production-grade conversational therapy AI built to demonstrate a full Generative AI Lead Engineer skillset. Multi-provider LLM backend with automatic fallback, 6-layer memory architecture, LangGraph multi-agent orchestration, and RAG pipeline grounded in DBT clinical content.

[GitHub](#) | Demo available upon request

- Multi-provider LLM abstraction layer (OpenAI, Anthropic, DeepSeek) with automatic fallback routing and token budget enforcement
- 8-node LangGraph agent graph with safety routing, crisis short-circuit, and full audit trail via routing_path
- 6-layer memory architecture: semantic cache, sliding-window history, two-tier working memory, episodic store, long-term gems (personal facts, sensitive disclosures, psych patterns), and per-turn special instructions
- RAG pipeline with LLM query rewriting, ChromaDB retrieval, score-threshold filtering, and DBT knowledge grounding
- GateAgent (5-classifier: safety, RAG, sensitive, personal_info, topic_shift) + AlphaAgent output validation (clinical claim, hallucination, low confidence)
- A/B testing framework (ExperimentRunner) with concurrent variant execution and /compare/vote endpoint
- Sensitive topic detection with persistent Redis flag across LangGraph invocations and careful post-session summarization
- Async FastAPI backend, structlog structured logging, Docker containerization, pydantic-settings config